

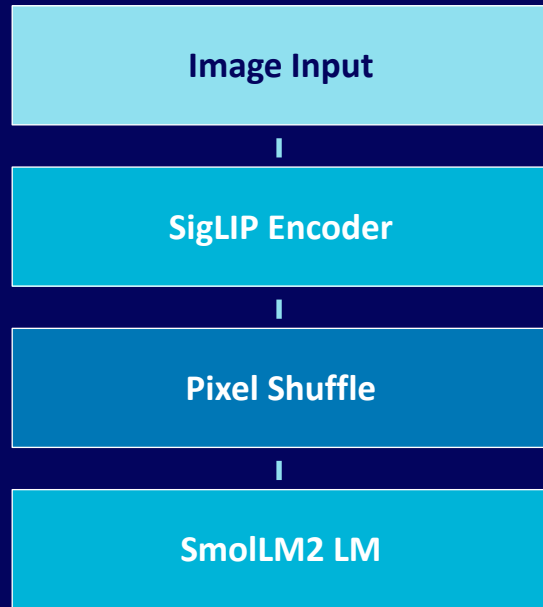
# SmolVLM: Redefining Small and Efficient Multimodal Models

Compact Vision-Language Models for Edge & On-Device AI

Hugging Face & Stanford University

arXiv:2504.05299

Architecture Overview



# Large VLMs Are Powerful but Impractical for Edge Deployment

## The Problem

- State-of-the-art VLMs require 27.7 GB+ GPU RAM
- Cloud dependency creates latency and privacy risks
- Mobile/edge devices have 4–16 GB total memory
- Inference costs prohibit consumer applications

Molmo-A1B-7B

**27.7 GB**

GPU RAM Required

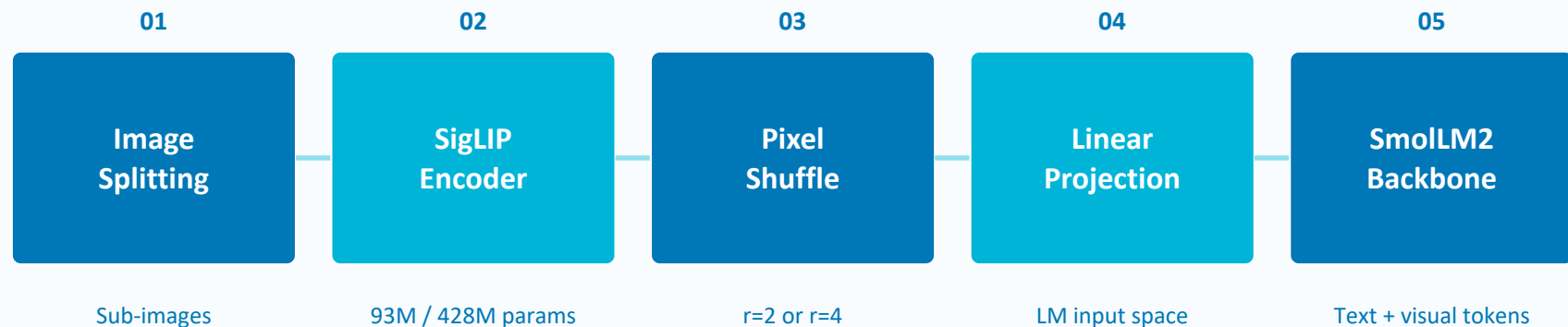
SmolVLM-256M

**0.8 GB**

GPU RAM Required

**34× smaller**

# SmolVLM Architecture: Image Splitting, Pixel Shuffle & SmolLM2



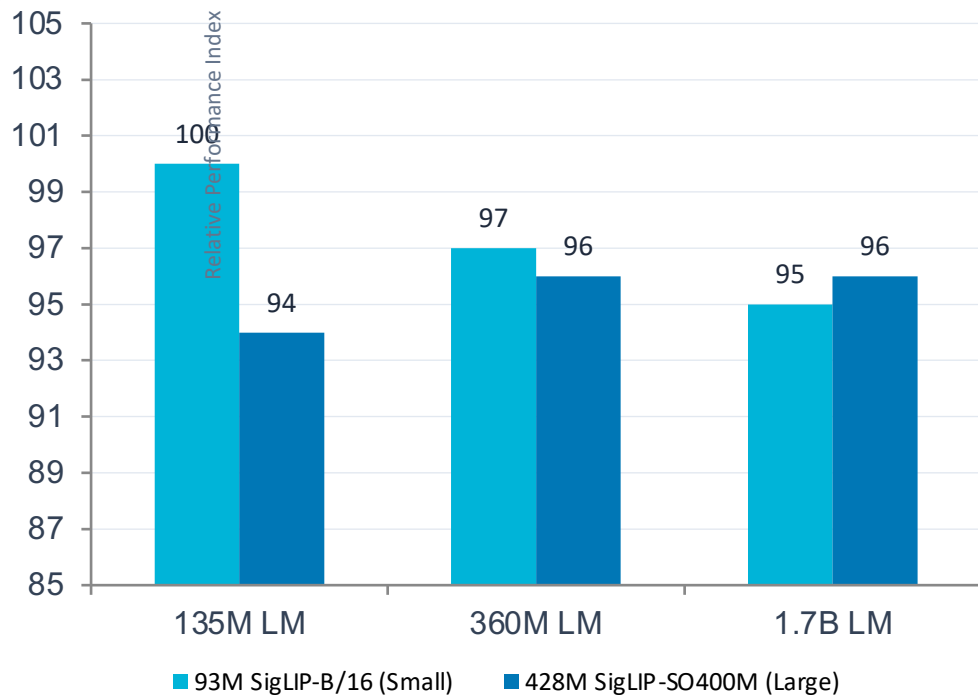
## Key Design Decisions

- Balanced encoder-LM compute allocation
- Aggressive pixel shuffle token compression
- Extended 8K–16K context window

## Model Family

- 256M params — 0.8 GB RAM
- 500M params — 1.2 GB RAM
- 2.2B params — 4.9 GB RAM

# Smaller Vision Encoders Outperform Larger Ones in Compact VLMs



## Key Finding

**93M encoder outperforms 428M at small LM scale**

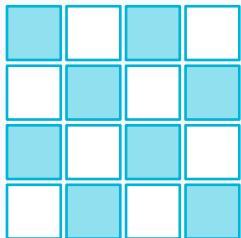
- Smaller encoder = less compute overhead
- Balanced encoder-LM sizing is critical
- Large encoder justified only at 1.7B+ LM

SigLIP-B/16: 93M params

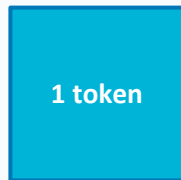
SigLIP-SO400M: 428M params

# Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Before Pixel Shuffle (r=4)



After Pixel Shuffle



16 tokens → 1 token

| Shuffle Factor $r$ | Token Reduction  | Best For                    |
|--------------------|------------------|-----------------------------|
| $r = 2$            | 4× fewer tokens  | Larger models ( $\geq 1B$ ) |
| $r = 4$            | 16× fewer tokens | Small models ( $< 500M$ )   |

## Why $r=4$ for Small Models?

16× fewer tokens reduces attention complexity dramatically

Enables longer context windows at same memory budget

Larger models already have capacity — less compression needed

**Token Reduction =  $r^2$**

$r=4 \rightarrow 16\times$  |  $r=2 \rightarrow 4\times$

# Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

## Context Extension Strategy

### Step 1: Baseline 2K Context

RoPE base = 10,000 | Limited image resolution

### Step 2: RoPE Base → 273,000

Enables position encoding at extended context lengths

### Step 3: Mixed Context Fine-tuning

Training on short + long context sequences simultaneously

### Result: Up to 16K Tokens Context

2.2B model: 16K | 256M/500M: 8K

## Impact

8x

Context Length Increase

2K → 16K

for 2.2B model

2K → 8K

for 256M/500M

Supports more sub-images per query —  
higher effective resolution

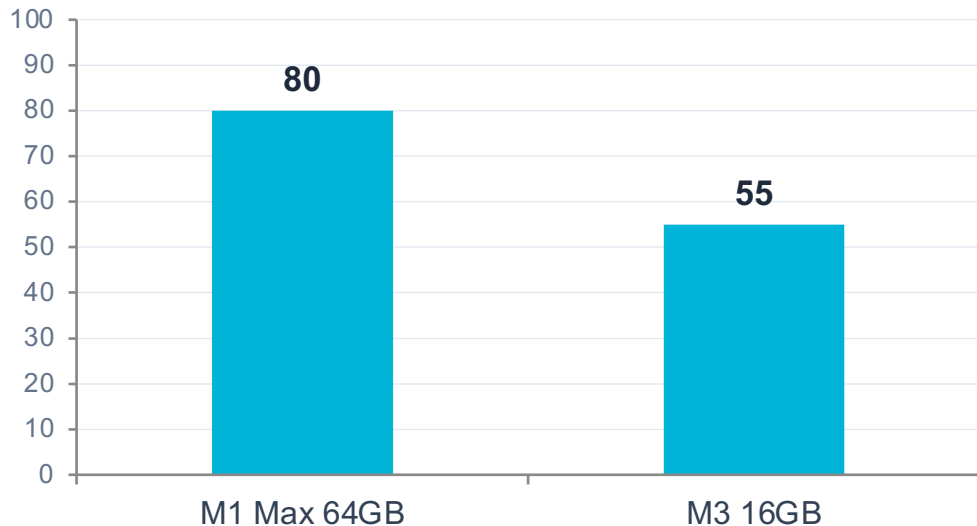
# SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

| Benchmark | 256M   | 500M   | 2.2B   | Best Efficient OS    |
|-----------|--------|--------|--------|----------------------|
| OCRBench  | 52.6%  | 61.0%  | 72.9%  | 54.7% (Molmo-A1B-7B) |
| AI2D      | 46.4%  | 59.2%  | 70.0%  | 71.0% (Molmo-A1B-7B) |
| ChartQA   | 55.6%  | 62.8%  | 68.7%  | 48.0% (Molmo-A1B-7B) |
| TextVQA   | 50.2%  | 60.2%  | 73.0%  | 61.5% (Molmo-A1B-7B) |
| DocVQA    | 58.3%  | 70.5%  | 80.0%  | 77.7% (Molmo-A1B-7B) |
| ScienceQA | 73.8%  | 80.0%  | 89.6%  | 87.5% (Molmo-A1B-7B) |
| MMMU      | 29.0%  | 33.7%  | 42.0%  | 33.9% (Molmo-A1B-7B) |
| MathVista | 35.9%  | 40.1%  | 51.5%  | 37.6% (Molmo-A1B-7B) |
| MMStar    | 34.6%  | 38.3%  | 46.0%  | 43.1% (Molmo-A1B-7B) |
| Video-MME | 33.7%  | 42.2%  | 52.1%  | 45.0% (InternVL2-2B) |
| MLVU      | 40.6%  | 47.3%  | 55.2%  | 48.2% (InternVL2-2B) |
| AVERAGE   | 44.0%  | 51.0%  | 59.8%  | —                    |
| RAM Usage | 0.8 GB | 1.2 GB | 4.9 GB | 27.7 GB (Molmo)      |

SmolVLM-256M surpasses 300× larger Idefics-80B · Only 0.8 GB RAM vs 27.7 GB for Molmo-A1B-7B

# Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

Throughput: Tokens/Second (Batch=1)



Batch=64: ~1,000 tok/s on Pro M1 Max — linear scaling with batch size

## HuggingSnap App

100% On-Device — No cloud required

Processes photos & videos locally

Private inference on iPhone hardware

~80

tok/s (M1 Max)

~55

tok/s (M3 16GB)



# Key Takeaways

**01**

## Balance Encoder-LM Compute

93M over 428M for small LMs

**02**

## Extend Context Length

2K  $\rightarrow$  8K–16K via RoPE base 273K

**03**

## Aggressive Token Compression

$r=4 \rightarrow 16\times$  token reduction

**04**

## Sub-1GB Inference

0.8 GB RAM on SmolVLM-256M

**05**

## Fully Open-Source

Models, data, code, HuggingSnap app